

growing hard disks and all—so why not store and/or watch movies and TV on them? That's where 3G and 4G and all come in; they enable everything on the gadget, just like the Internet connection will (and in some places, does) enable everything on the PC. The Gadget is surpassing the laptop.

But human nature is never content. (A silly truism, because human nature *means* "never content.") So, well, who wants to watch stuff on a teeny-weeny 3-inch screen? Enter the holographic display. The technology could soon be mainstream: the Gadget might well project movies into the air over a large area. And as for the input, who wants to type or write onto a 3-inch pad? The answer is the projection keyboard, which "projects" keys onto any surface you place the Gadget on. And hey, if you can do all this on the Gadget, can real 3D games on a 3D surface be far behind?

Getting back to the Internet, what we've been seeing is expanded use. Blogging comes to mind first: as the world increasingly deals with information in byte-sized chunks, people increasingly turned towards bite-sized write-ups called blog posts instead of long, boring Web pages to navigate through. Then there's MMORPGs: that's real virtual reality. People in countries where the bandwidth lies live, sleep, work, play, and even mate online. They can't eat online—yet; although at least one MMORPG allows you to order pizza through its interface. When will that promised food-pill arrive? No, we're beyond kidding now: nanotech might well produce that staple of science fiction.

While on the subject of the Internet, we must talk about the

real problem of our age—the clichéd "information overload." There are two problems here: storing it all, and retrieving it all. As for the retrieval, trust all the search companies out there—and, going a level beyond, the data mining companies. Yes, data mining and text mining will become more important. We want only what we need: data mining will provide precisely that, patterns gleaned from heaps of information. Text mining will enable us to read summaries of articles, the main points of a news item, and so on. Like we said, we like everything in bite-sized chunks these days.

Coming to the storage, just make the hard disks larger! Did we say "hard disks"? No, no, it won't be hard disks. There's already Flash memory replacing it—initially for faster bootups—and then there will be holographic storage (due out commercially this year, by the way), and ultimately carbon nanotube storage. A pinhead of nanotubes will hold terabytes.

Everything else is moving on, but when it comes to the interfacing with our Gadgets and our PCs, we're stuck with mouse, keyboard, and stylus. Will AI please come to the rescue? How soon will we be able to just tell our display device to play a certain movie?

And when you talk about interaction, think of pervasive computing: already, researchers are talking about it. Check out www.pervasive.dk: "Pervasive computing is the next generation computing environments with information and communication technology everywhere, for everyone, at all times. Information and communication technology will be an integrated part of our environments: from toys, milk cartons and desktops to cars, factories and

whole city areas—with integrated processors, sensors, and actuators connected via high-speed networks, and combined with new visualisation devices—ranging from projections directly into the eye to large panorama displays."

"Projections directly into the eye"? Now that's an idea, come to think of it!

The decade so far has been, and the years to come will be, not so much about new technologies such as perpendicular recording for hard disks. It's more about people generally having come to believe—perhaps rightly so—that technology can give us whatever we want.

On-the-fly. Instant. Anyplace. Anytime. More. Even more. We want. It's moving fast. It's getting faster. It's getting smaller. Into our palms. Under our skin. Everywhere around us. Possibly inside us. God is now well and truly the Great Geek in the Sky—and what we want, as of this decade and in the decades to come, is for *all our material desires* to be fulfillable anytime, anyplace, right next to us, on demand. ■

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The Gadget

It's the definitive development of this decade thus far: the portable Gadget. It was a cell phone at first, and at some point, when both cell phones and PDAs were being used, it became obvious that they could be combined. Some called the new gadget the smartphone. Then the big idea: if everyone's got this in their pockets, why not make it more capable? They added a digital music player. They added FM radio. They added faster processors and more memory. They Web-enabled it. They added a camera, so you could click snaps and instantly send them off to someone via e-mail. ("Look! Here's where I am!") And the cameras are piling on megapixels, the network is gaining in speed, the processors are getting better, and all the rest of it.

The Gadget, as we're calling it, *represents* convergence.

But it's not just about convergence, or even about convenience. It goes beyond: we want to break free of the shackles of time and space, as it were.

